

GeM BidGuard — AI-Powered Integrated Bid Compliance Verification Platform

Ministry of Petroleum & Natural Gas · Chennai Petroleum Corporation Limited (CPCL)

REQUIREMENT-TO-EVIDENCE COMPLIANCE ENGINE

1. Detailed Explanation of the Proposed Solution

GeM BidGuard is an AI-assisted platform that verifies the statutory, regulatory and eligibility compliance of every vendor bidding on the Government e-Marketplace. It replaces the officer's manual portal-by-portal, document-by-document check with a single automated pass that returns a scored, evidence-backed assessment — while leaving the qualification decision entirely with the Procurement Officer.

The platform is built on a **Requirement-to-Evidence** principle. Each tender is decomposed into a structured compliance checklist, and every requirement on that checklist is resolved against evidence drawn from two independent streams: **live government portal records** and the **bidder's submitted documents**. No requirement is marked compliant without a citable source.



Multi-portal evidence acquisition

Thirteen government sources are integrated behind a single uniform connector interface — Udyam/MSME, GSTN (registration *and* return-filing history), PAN/Income Tax, MCA21, EPFO, ESIC, Startup India (DPIIT), NSIC, DigiLocker, DPIIT local-content, BIS, OEM authorization, and an aggregated blacklist/debarment scan spanning GeM, CPPP, MoPNG and individual CPSE banned-vendor lists. Every connector returns an identical envelope — status, payload, timestamp and a citable evidence reference — so a source can be added or upgraded without any downstream change.

Requirement-level verification and reporting

For every requirement the platform records the **compliance status**, the **source of evidence**, the **evidence reference**, the **reason for the result** and the **weight** it carries in the final score. The buyer receives a requirement-by-requirement report rather than a single opaque verdict, and every line in that report can be independently re-verified by a human.

Architectural principle: rules decide, AI explains

A deterministic rule engine produces every compliance status and the numeric score. The language model is confined to two tasks it is genuinely better at — reading unstructured documents, and turning the finding set into a briefing an officer can place on file. **The model cannot alter a status, a score or a recommendation.** This is a direct consequence of the problem statement: a decision that must be auditable and reserved to a human officer cannot depend on a non-reproducible output.

2. How It Addresses the Problem

PROBLEM TODAY	HOW GEM BIDGUARD RESOLVES IT
Manual portal-hopping	A 40-bidder tender requires roughly 500 manual lookups across a dozen captcha-protected portals. The platform reduces this to a single action, targeting the 60–80% effort reduction the problem statement specifies.
Registration checked, compliance missed	A GSTIN can read "Active" while the firm has filed no return for six months. The platform verifies filing behaviour , not merely registration — consecutive GSTR-1/3B defaults, EPFO and ESIC arrears, Section 206AB status, and Udyam suspension for non-update.
No cross-checking between sources	The same fact — legal name, PAN, registered address, incorporation date — is pulled from every portal that reports it and reconciled. Divergent PIN codes across Udyam and GST surface immediately.

Missing, expired and forged documents	Mandatory-document completeness is computed from the tender's own requirements. DigiLocker-issued documents are distinguished from uploaded PDFs, which can be altered. Certificate validity is checked against the bid date.
Every bid reviewed at equal depth	A compliance score and risk band triage the queue, so officer attention goes to contradictory, expired or ambiguous evidence instead of to files that are demonstrably clean.
Inconsistent and unauditable outcomes	Identical rules apply to every bidder across every CPSE. Every portal query, rule evaluation, score computation and officer action is written to an append-only trail with evidence references.
Accountability of the decision	The officer receives a recommendation, the reasoning and the underlying evidence — and records the decision, with remarks, into the same audit trail. Qualification and disqualification remain entirely with the Procurement Officer.

3. Innovation and Uniqueness of the Solution

Relationship-aware debarment detection — the core innovation

Debarment is enforced today by searching a banned-vendor list *by company name*. A debarred promoter defeats that check by incorporating a new company. GeM BidGuard resolves debarment through the **MCA21 director graph**: it matches bidders to debarred entities by shared DIN, shared registered address and incorporation timing. In validation, a bidder that is clean on every direct check — active Udyam, current GST filings, valid PAN, absent from every blacklist — was linked through its Managing Director's DIN to an entity debarred by IOCL until 2027, operating from the same registered address, and incorporated 139 days after that debarment took effect. **A name-based search returns nothing. This class of evasion is invisible to manual verification and is not addressed by any existing tool.**

Explainable, legally defensible verdicts

Separating a deterministic decision layer from a generative explanation layer means every status is reproducible from the inputs alone. Under Indian administrative law an officer must apply their mind to a decision; "the model scored the bidder 34" is not a defensible ground if a disqualification is challenged. This architecture gives the officer a reason and an evidence reference for every finding.

The score is never silently adjusted

A gating failure — debarment, invalid PAN, cancelled GSTIN, suspended Udyam on an MSE-reserved tender — overrides the risk band to CRITICAL and is displayed *alongside* the documentary score, never folded into it. A bidder can read 85/100 on paperwork and still be CRITICAL. The officer sees both numbers and the reason for the divergence, rather than an unexplained low score.

Audit-ready by construction

The requirement, its evidence, the source location, the reasoning, the score contribution and the officer's final action are maintained as a single linked record with hashed evidence references — satisfying the problem statement's requirement for complete auditability and traceability, and standardising verification across CPSEs.

False-positive suppression

Naive cross-checking produces noise that trains officers to ignore the tool. The reconciliation engine clears formatting variants ("Pvt Ltd" / "Private Limited") automatically, and applies entity-type awareness — a proprietorship's PAN is held in the proprietor's personal name, so it is **not** reported as a name mismatch. Only genuine disagreements reach the officer.

Deployment-ready integration interface

Government portals expose heterogeneous and access-restricted interfaces — GST requires a GSP licence, DigiLocker requires partner onboarding. Every connector is therefore written against one uniform envelope, so authorised credentials can be introduced source-by-source at deployment without redesign. The compliance logic is independent of how any single source is reached.

Working prototype: Next.js platform with a headless verification engine, 13-source connector layer, 16-check compliance engine, cross-source reconciliation, weighted scoring with gating overrides, AI recommendation with deterministic fallback, document extraction, and a procurement-officer dashboard with a full audit trail. Validated against four bidder profiles exercising the clean, mismatch, statutory-lapse and debarment-linkage paths. ·

Decision-support tool. Qualification and disqualification remain with the Procurement Officer.